

MaskCircuit: 3D Circuits with Acrylic Sheets and Laser Cutting

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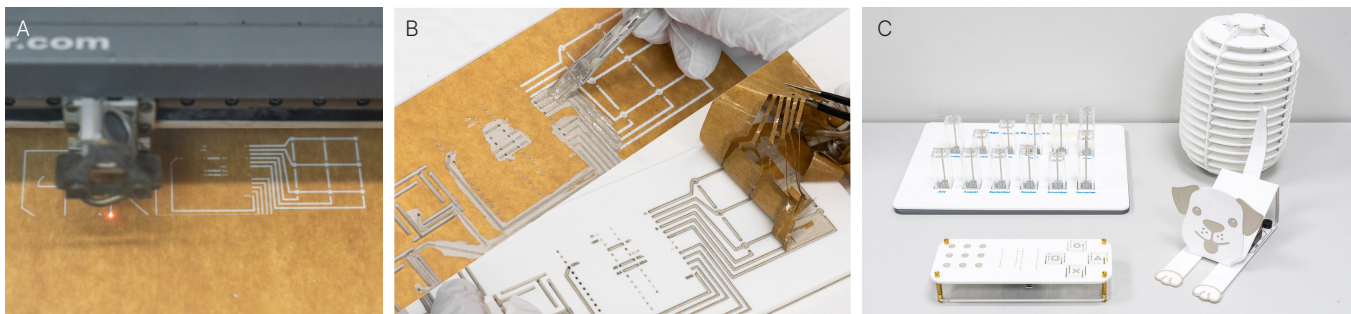


Figure 1: A: Laser engraving traces onto a masked acrylic sheet. B: Applying conductive paint and removing the mask. C: Interactive applications fabricated with *MaskCircuit*

ABSTRACT

Laser cutting is an accessible fabrication technique for designers to create supporting structures for tangible interfaces and electronic devices. However, it is challenging to integrate electronics into laser-cut prototypes due to bulky prefabricated components and wiring. To address these challenges, we propose *MaskCircuit*, a maker-friendly fabrication approach for 3D circuits based on commonly used laser cutters and acrylic sheets. *MaskCircuit* leverages the protective film of acrylic sheets as a mask. Circuits are created on acrylic sheets by laser engraving traces into the masked acrylic sheets, manually applying conductive paint into traces, and removing the mask to reveal inlaid conductive traces. We also explored circuit features supported by *MaskCircuit*, including double-layer circuit fabrication, mounting electronic components, and creating 3D circuits by deforming or assembling acrylic sheets with circuits. We demonstrate several applications to showcase the capabilities of the *MaskCircuit* approach.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

Digital Fabrication, Laser Cutting, Tangible Interfaces

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1 INTRODUCTION

Laser cutting is a fabrication technique that is accessible to many designers, researchers, and individual makers. This is largely due to the wide availability of CO₂ laser cutters in makerspaces and design workshops. Laser cutters work easily with existing design software, are straightforward to use, operate on a wide range of materials, and are capable of precise and rapid fabrication.

With regard to HCI, laser cutting is often used to make supporting structures (e.g. enclosures) for tangible interfaces and electronic devices by fabricating 2D pieces that are assembled into 3D objects [2]. Fabricating laser cut structures to support electronic components for a tangible interface is a complex design task—particularly during early prototyping. Designers need to design around bulky prefabricated components such as breadboards,

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jumper wires, and breakout components while maintaining good circuit connections. Alternatively, designers might outsource circuit fabrication to printed circuit boards (PCB) for more compact circuits. However, PCB design and fabrication require specialized software expertise while the production process is often time-consuming. Furthermore, the rigid shape of PCB poses similar integration challenges for laser cutting.

To address the challenges of integrating electronics into laser cut prototypes, we developed *MaskCircuit*: a maker-friendly fabrication approach for 3D circuits based solely on commercially available laser cutters and acrylic sheets. *MaskCircuit* enables designers to easily integrate circuits into laser cut assemblies through an unmodified laser cutter and acrylic sheets.

The key feature of this approach is our new use of the protective film that typically covers both faces of an off-the-shelf acrylic sheet. *MaskCircuit* extends the use of this film as a circuit mask (Figure 1):

- (1) Circuit traces are laser engraved onto the masked acrylic sheet, selectively removing the mask and engraving into the underlying acrylic based on the circuit design. (Figure 1A)
- (2) Conductive paint is manually applied into the engraved traces. (Figure 1B)
- (3) The mask is peeled off, leaving a clean acrylic sheet with inlaid conductive traces. (Figure 1B)

MaskCircuit also supports common PCB features. Double layer circuits can be created by engraving both sides of the acrylic sheet and connecting the traces with laser cut holes that are filled with conductive paint (i.e. vias). Through-hole or surface-mount electronic components can be attached through laser-cut holes or laser-engraved pads respectively, and attached through conductive epoxy. In addition, the acrylic sheet with inlaid circuit can be deformed with heat, or assembled with other acrylic sheets to create 3D circuits.

MaskCircuit integrates circuit fabrication into the typical process of designing and fabricating a laser cutter assembly with acrylic sheets, while the mask engraving process enables us to create detailed circuits directly on the acrylic sheet itself through a manual process without the need for specialized tools. In the following sections, we will discuss the related work that inspired us, detail the *MaskCircuit* approach, as well as demonstrate several applications that showcase the capabilities of the approach.

2 RELATED WORK

Our work relates to the research around fabricating circuits with the laser cutter as well as building 3D circuits.

2.1 Fabricating Electronic Circuits with the Laser Cutter

Researchers have leveraged laser cutters in different ways for fabricating circuits. For instance, laser cutters can be used to cut masks and stencils to facilitate the precise application of conductive material [10, 23]. Laser cutters can also be used to directly cut conductive materials and circuit substrates to fabricate circuits. Such research expands the physical materiality of electronic circuits. For example, *LASEC* [3] demonstrates deformable and stretchable plastic circuits, *Sensing Kirigami* [26] demonstrates paper circuits capable of sensing folding and bending, and *CircWood* [5] demonstrates

wood circuits created by selectively carbonizing wood sheets with a laser cutter. These examples leverage a commodity CO_2 laser cutter to fabricate circuits. Besides that, *FiberCuit* [25] used a fiber laser engraver machine to laser cut copper-based composites to form fine-pitch circuits. Researchers have also developed new machines based on laser cutting to support fabricating electronic circuits and devices. *LaserFactory* [11] uses a modified laser cutter to print circuit traces, pick and place components, as well as solder electrical connections—all within one machine.

The approach we propose hones in on acrylic sheets. We focus on supporting designers and makers to fabricate circuits on top of this common material using tools easily found in design workshops and maker spaces. Rather than developing new software or machine capabilities, we focus on the existing materiality of commodity acrylic sheets—leveraging the protective film of this material as a mask to efficiently incorporate intricate conductive traces into acrylic sheets.

2.2 Fabricating 3D Electronic Circuits

HCI Researchers have developed different ways to fabricate 3D circuits to facilitate interactivity for non-planar and 3D tangible interfaces. One approach is to fabricate conductive traces on a 2D substrate and transform or deform it into a 3D structure. For instance, inkjet printed circuits can be folded into 3D electronics for sensing [7, 8, 12–14], displays [8, 12, 13], or entire wearables [24]. *Microcontrollers as Material* [9] demonstrated drawing conductive ink on paper while *Foldable Printed Circuit Boards on Paper Substrates* [18] deposited metal on paper directly to build foldable 3D circuits.

Another approach is to embed 3D conductive traces directly into objects with 3D printing. For instance, *Oh, Snap!* [16], *Flexibles* [17], and *Capricate* [15] printed conductive traces with conductive filament inside a plastic part for capacitive touch sensing. Similarly, *FiberWire* [19] embedded conductive carbon fiber into parts during 3D printing to build both interactive and mechanically strong electronic devices. Besides 3D printing conductive circuits, *SurfCuit* [20] used copper tape and copper tubes to instrument 3D printed objects with circuits.

In addition, rather than fabricating 3D circuits directly, 2D circuits can be formed into 3D objects after fabrication. *MorphSensor* [27] attached 2D ink-jet printed circuits onto the surface of 3D objects with double-side tape. *Thermoformed Circuit Boards* [4] transforms flat PLA parts embedded with conductive traces into 3D circuit boards by heating and shaping the part by hand. *MorphingCircuit* [22] uses 4D printing to transform a 2D circuit into a preprogrammed 3D shape.

Inspired by these methods, we take advantage of the characteristics of acrylic and common maker practices with this material and propose different ways of creating 3D circuits with acrylic sheets. Like *Thermoformed Circuit Boards* [4], acrylic sheets with embedded circuits can be thermoformed by hand into 3D parts. In addition, we demonstrate 3D circuits composed by joining different 2D acrylic parts using joinery commonly used for laser cut assemblies.

3 APPROACH

We detail the specific design and fabrication steps for *MaskCircuit*:

3.1 Circuit

Our approach relies on an unmodified laser cutter and designers can use vector drawing tools (e.g. Adobe Illustrator, Inkscape) to create the circuit design file.

We conducted initial tests to evaluate the circuit design parameters possible with our approach. We detail our findings in Figure 2.

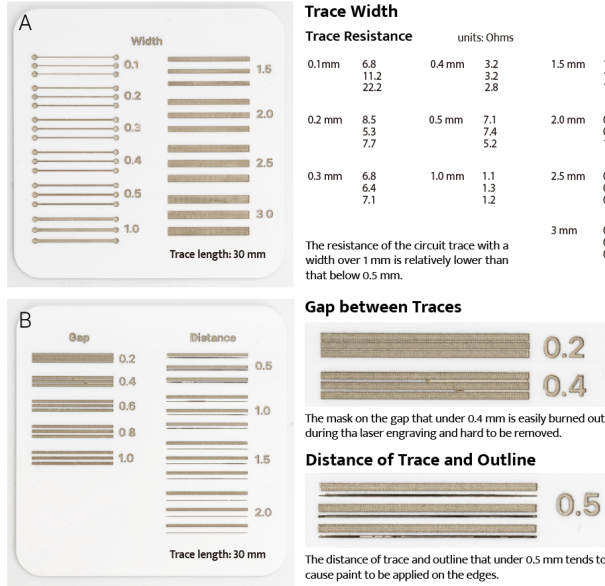


Figure 2: A: Electrical resistance measurements for conductive trace samples of varying widths fabricated with MaskCircuit. B: Evaluating the resolution of gaps between traces, as well as between trace-outline for MaskCircuit.

3.2 Laser Cutter Processes

3.2.1 Masking. We used the POLY-A cast acrylic sheets that come covered with a protective film on both sides. This stock protective film is a suitable mask for laser engraving and applying conductive paint. Bare acrylic sheets can be covered with masking tape (e.g. “painters tape”).

3.2.2 Laser Engrave Circuit Traces. In this work, we used a commercial CO₂ laser machine (Epilog Helix 30 Watt) to engrave the circuit traces (settings: power: 19.5 W, speed: 338mm/s). As Figure 3A shows, the engraved surface is recessed and has a rough surface finish. This ensures that the conductive paint adheres well to the acrylic sheet.

3.2.3 Laser Cut Acrylic Sheet. Physical features, such as the shape of the part, are then laser cut onto the engraved acrylic sheet (Figure 3B). We perform this operation without removing the acrylic sheet from the machine after engraving to ensure accurate registration of circuit details and physical features.

3.2.4 Double-Layer Circuits. Double-layer circuits can be created with this approach with slight adjustments to the sequence of steps. First, the top layer circuit is laser engraved on the masked acrylic sheet. Second, via holes (Figure 3B) as well as physical features, including the part outline, are laser cut. Third, the acrylic part is flipped (Figure 3C) and the bottom layer circuit is engraved

(Figure 3D). We use a jig in the laser cutter with the form of a mirrored outline to ensure proper registration of the flipped part.

3.3 Conductive Paint Application

3.3.1 Brushing Conductive paint. We manually brush conductive paint into the engraved traces (Figure 3E). In this work, we use an acrylic-based silver conductive paint (MG Chemicals 842AR) as it has low resistivity ($1.0 \times 10^{-4} \Omega \cdot \text{cm}$). We experimented with different brushes and found acrylic paint brushes to be suitable for applying the conductive paint. The conductive paint we use dries rapidly which might result in uneven thickness of the paint coat. We addressed this by thinning the conductive paint with thinner in a 1:1 ratio (MG 4351 Thinner 1) before applying it. The thinned paint also flows more easily into small traces and holes.

3.3.2 Remove Mask. Once the conductive paint is dry to touch, we can peel off the mask to remove excess paint—revealing the clean conductive traces of the circuit (Figure 3F). Thin masks that are difficult to peel by hand can be easily removed with a scraper as the conductive traces are recessed and protected from surface abrasion.

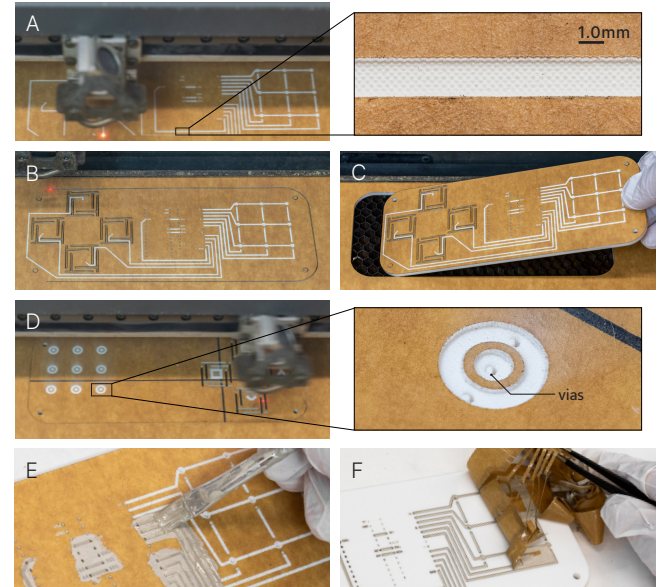


Figure 3: Fabrication process for MaskCircuit: A: Laser engraving circuit traces onto the acrylic sheet with surface detail showing a recessed and textured surface. B: Laser cutting physical features without moving the acrylic sheet. C: Flipping the laser cut part in preparation for double-layer circuits. D: Laser engraving circuit traces on the bottom layer of flipped laser cut part with via holes for double-side circuit traces connection. E: Applying conductive paint into traces and vias with a brush. F: Peeling off the mask with excess conductive paint on.

3.4 Circuit Components and Assembly

We explored incorporating electronic components and sensing traces into the circuits fabricated with our approach:

3.4.1 Sensing Circuits. Conductive traces fabricated with MaskCircuit can be easily instrumented to detect touch or bare skin proximity based on capacitive sensing. In this work, we fabricated touch sensing traces (Figure 4A) with the Capacitive Sensing Library of Arduino [1] and MPR121 (through the Adafruit 12-Key Capacitive

Touch Sensor Breakout). Fabricated traces can also be used to detect the bending of the acrylic part (Figure 4B), since the resistance of conductive traces will increase or decrease during bending [21]. We fabricated bend sensing traces with Arduino and a voltage divider circuit.

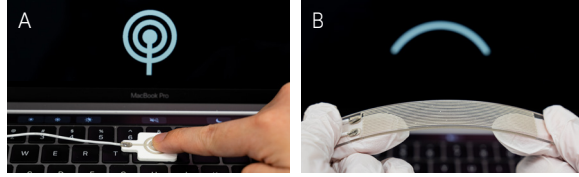


Figure 4: A: Capacitive touch sensor. B: Resistive bend sensor.

3.4.2 Attaching Electronic Components. We attached off-the-shelf electronic components onto the conductive traces with silver conductive epoxy adhesive (MG 8331, resistivity: $7.0 \times 10^{-3} \Omega \cdot \text{cm}$). The conductive epoxy we used cures at room temperature within 24 hours, or within 15 minutes at 65°C . We explored attaching two main types of electronic components: surface-mount devices (SMD) and through-hole devices (THD) (Figure 5). For SMD components, we place the component on the surface and inject conductive epoxy with a syringe between the component's leads and the conductive trace (Figure 5B). For THD components, we insert the leads of the component into the laser cut holes, and inject the epoxy onto the connection point between the leads and conductive trace (Figure 5D).

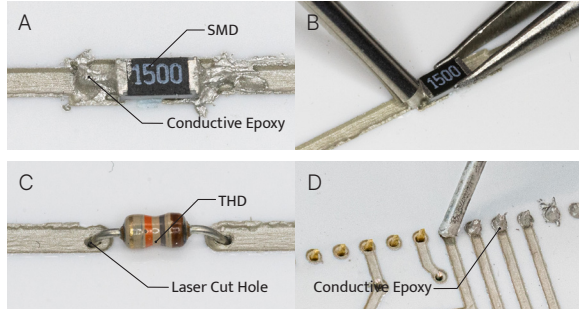


Figure 5: A, B: SMD components can be attached on the surface by injecting conductive epoxy between the component's leads and the conductive trace. C, D: THD components can be inserted into conductive holes and attached by injecting conductive epoxy into the connection point.

3.5 Forming and Assembling Acrylic Parts

3.5.1 Deform Acrylic Components. Acrylic sheets are easily deformed when heated to 140°C . Conventionally, designers heat up acrylic sheets through tools like the hot air gun, carbon fiber heater, or oven. In this work, we explored heating and deforming acrylic parts with conductive traces and electronic components applied (Figure 6). This thermoforming operation enables us to create 3D circuits through common makerspace tools.

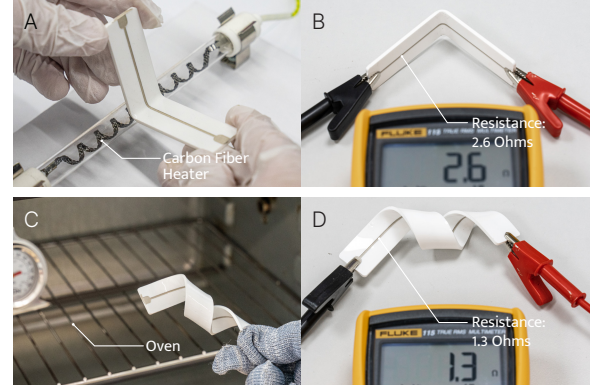


Figure 6: A, B: Bending an acrylic circuit heated with a carbon fiber heater. C, D: Twisting an acrylic circuit that was heated with an oven.

3.5.2 Joining Acrylic Parts. Joining acrylic parts is another common method to build 3D objects with flat acrylic sheets. We were inspired by various joining techniques for acrylic parts commonly used in the maker community. In our work, we explored four methods to connect circuits between different acrylic parts (Figure 7). This enables designers to construct both the electronic system and structure of the physical interface simultaneously.

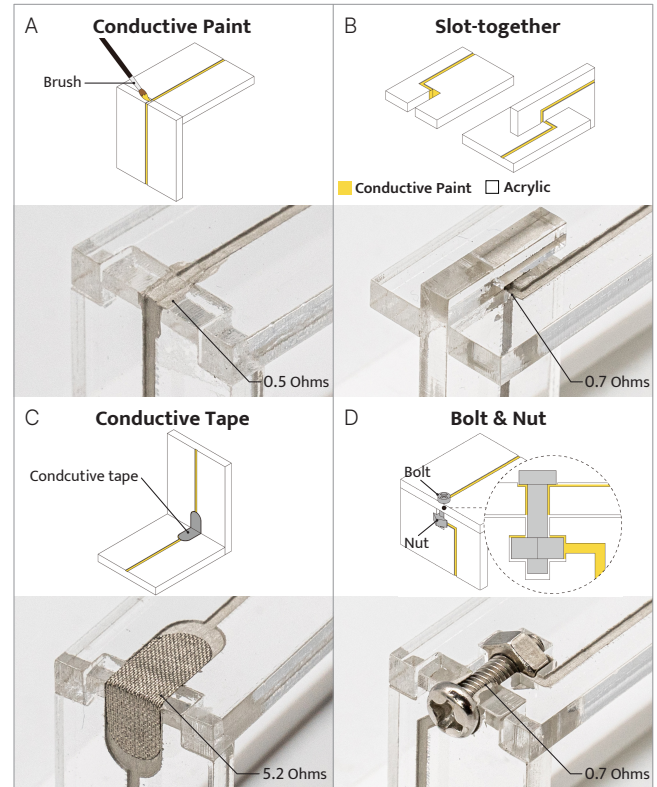


Figure 7: Four methods to connect circuits across common laser cut acrylic joints: (1) Brushing conductive paint across the joint. (2) Applying conductive paint onto the inner surface of slots that come into contact when joined. (3) Using conductive tape to connect circuits between acrylic parts. (4) Leveraging the conductivity of bolts and nuts to connect circuits across acrylic parts joined with a "T-slot".

4 APPLICATIONS

We used *MaskCircuit* to build a range of interactive applications. These applications demonstrate how designers might use this approach, along with other making processes, to create digital devices with acrylic sheets and laser cutting.

4.1 TUI: Handheld Controller

Laser cutting is commonly used to build enclosures for electronics. Here we showcase a controller with integrated circuits on the acrylic enclosure (Figure 8A). The controller features a touch panel and four push buttons (Figure 8B) that detect finger touch with the MPR121 touch sensing chip and an Arduino Nano microcontroller (Figure 8C). We took advantage of *MaskCircuit* to design a double-sided circuit. The double-sided circuit hides electronic components and circuit traces at the back of the interface, while exposing only the touch points at the front; resulting in a neat and clear user interface.

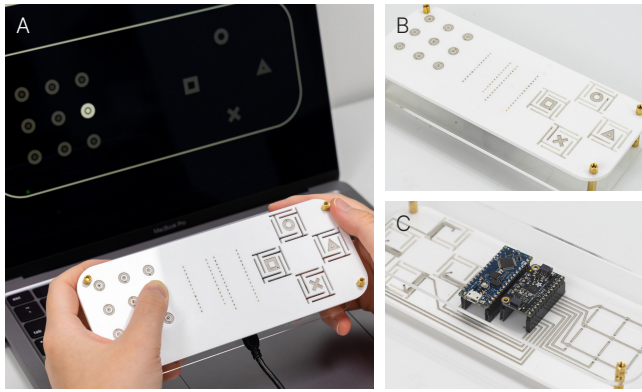


Figure 8: A, B: Handheld Controller with nine touch pads and four buttons. C: The bottom view of the controller with capacitive sensing breakout and microcontroller.

4.2 Data Physicalization: Rainfall Bar Chart

Clear acrylic sheets have excellent light transmission properties. We leveraged this material quality to build a data physicalization that shows average monthly rainfall in Singapore (Figure 9C). Each month is represented by an acrylic box that has a height relative to the rainfall of the respective month; much like a 3D bar chart (Figure 9A). These boxes are arranged on a touch screen tablet device which provides more information on the display.

We were inspired by *ExtensionSticker* [6] and incorporated conductive traces on the acrylic boxes that extend the touch detection of the screen to the top of each box (Figure 9B). This enables the system to detect when the user is interacting with a physical data point—and provide feedback such as by rendering more details about the data point, or by changing the color of the data point. Colors on the display are easily transmitted throughout the surface of the acrylic boxes to change the appearance of the physical data points as well.

4.3 Lifestyle Product: “Empty” Lamp

The slot-together joint is widely used in laser cut assemblies to create intricate 3D structures. We leveraged this joint to build a

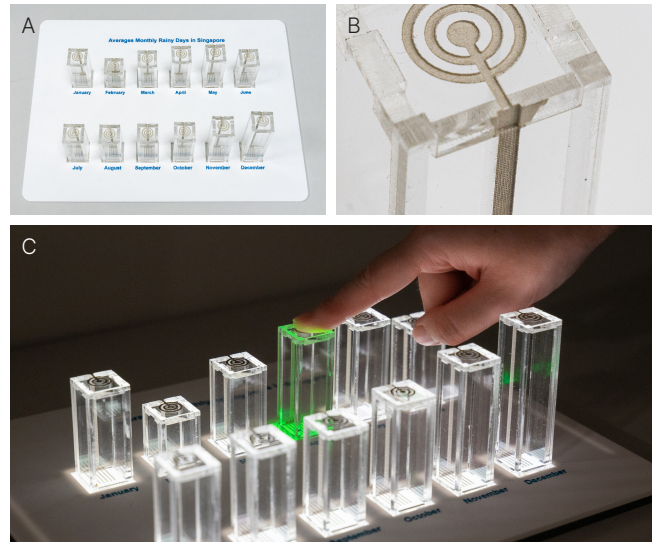


Figure 9: A: Different heights of acrylic bars show average monthly rainfall. B: Detail of the circuit connection with conductive paint running across different acrylic pieces. C: Data physicalization of rainfall with acrylic bars and a touch screen tablet device. The clear acrylic bars light up in a different color when users touch them.

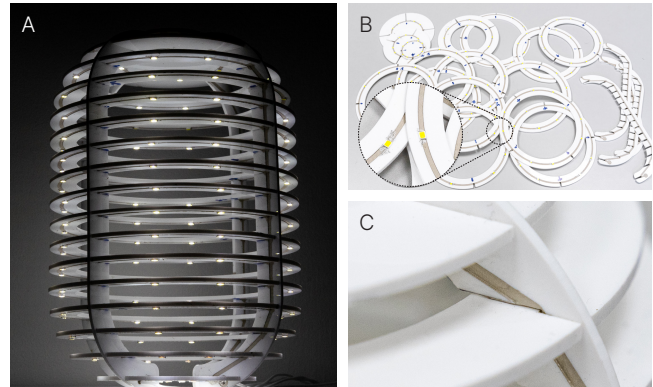


Figure 10: A: The “Empty” Lamp in use. B: Lamp parts with conductive traces and surface-mounted LEDs. C: Detail of the slot joint used to assemble the acrylic parts and connect circuits throughout the device.

lampshade that lights up without a conventional bulb (Figure 10A). We achieved this by mounting SMD LEDs onto acrylic rings that are held in place with vertical slot together pieces (Figure 10B). Conductive traces on the rings are electrically coupled to conductive traces on the vertical pieces through the slot-together joint (Figure 10C). This results in a visually “empty” lighting device without internal wires.

4.4 Toy Design: Interactive Toy Dog

We fabricated an interactive toy dog by thermoforming a single acrylic sheet with an embedded interactive circuit (Figure 11A). Touch sensing points and a bend sensor (Figure 11C) were incorporated into different parts of the toy. Audio and haptic feedback is provided during interaction via a piezo-buzzer and a vibration motor (Figure 11B). To form the 3D shape of the toy, we used a

carbon fiber heater to heat the acrylic part with electronics already mounted, and manually bent the part (Figure 11D).

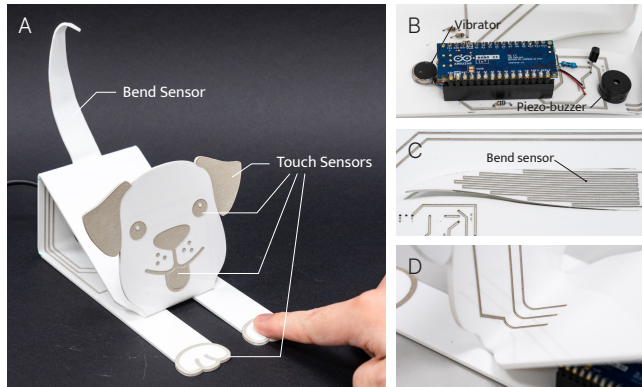


Figure 11: A: Interactive toy dog fabricated from a single sheet of acrylic. B: Mounted electronic components. C: Bend sensor before thermoforming. D: Detail of circuit running across a bend.

5 FUTURE WORK

Compared to conventional metal wire circuits or printed circuit boards, *MaskCircuit* traces are less uniform in terms of electrical resistance due to the manual painting process. We plan to systematically evaluate the electrical resistance of different trace design parameters and different paint application methods. From this technical evaluation, we aim to characterize the electrical properties of circuits fabricated with *MaskCircuit* to inform other researchers and designers on the limitations of this approach.

Our current approach enables designers to use familiar digital drawing tools to create circuit designs on top of laser cut acrylic sheets. However, we acknowledge that this task is challenging for novice designers and makers—particularly when designing traces to work with commercial electronic components, or designing double-sided circuits. We plan to develop a design editor to support designers in creating circuits fabricated with *MaskCircuit*.

Furthermore, we plan to run design workshops with other designers and makers to explore how they might use *MaskCircuit* to build interactive applications. These workshops will reveal the perceived capabilities and limitations of the approach and we intend to use these observations to refine *MaskCircuit*. Ultimately, we hope to use to document and publish a guide for *MaskCircuit* for designers and makers to refer to, including a repository of example applications from the community.

ACKNOWLEDGMENTS

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